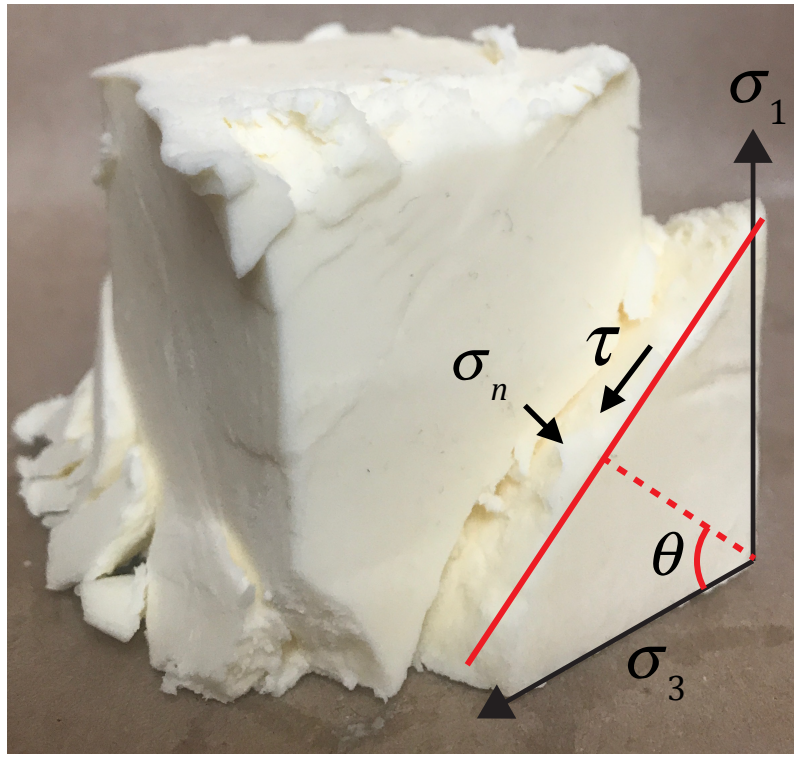
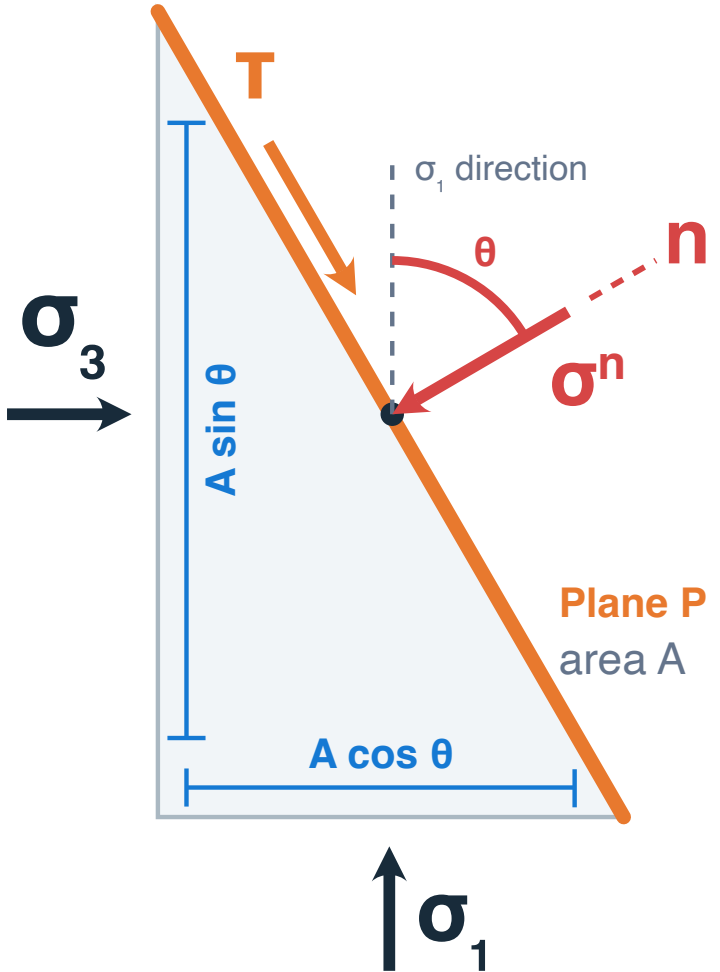
Force balance on an inclined plane

Mohr circle analysis

Graphical two-dimensional representation of the stress components acting on a plane of arbitrary orientation passing through a point.



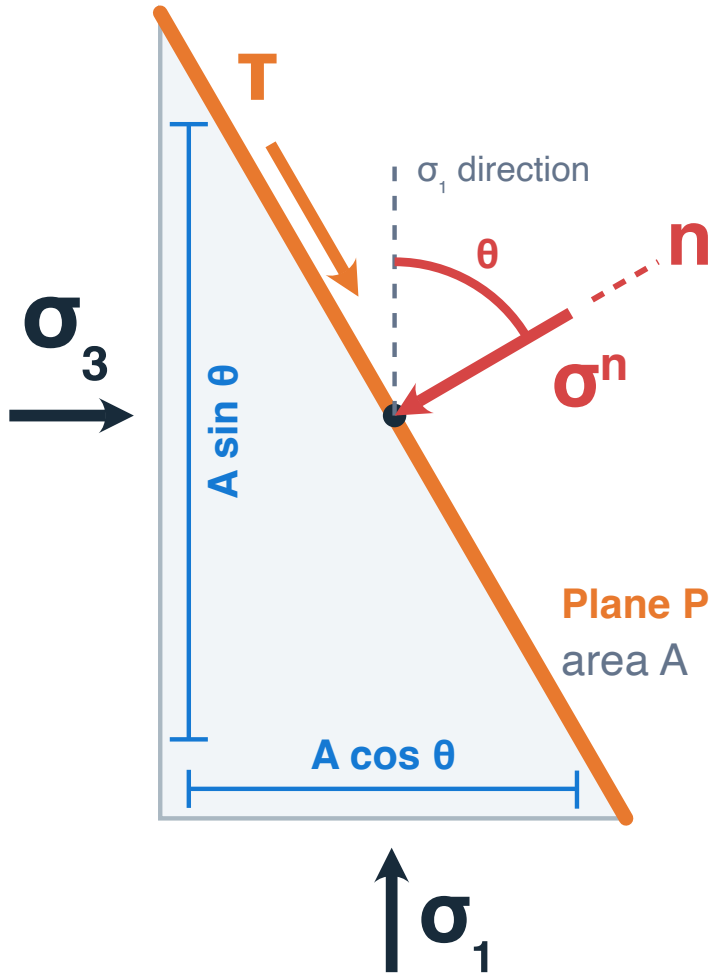
Neglecting the intermediate principal stress



Force balance on an inclined plane

Mohr circle analysis

Graphical two-dimensional representation of the stress components acting on a plane of arbitrary orientation passing through a point.



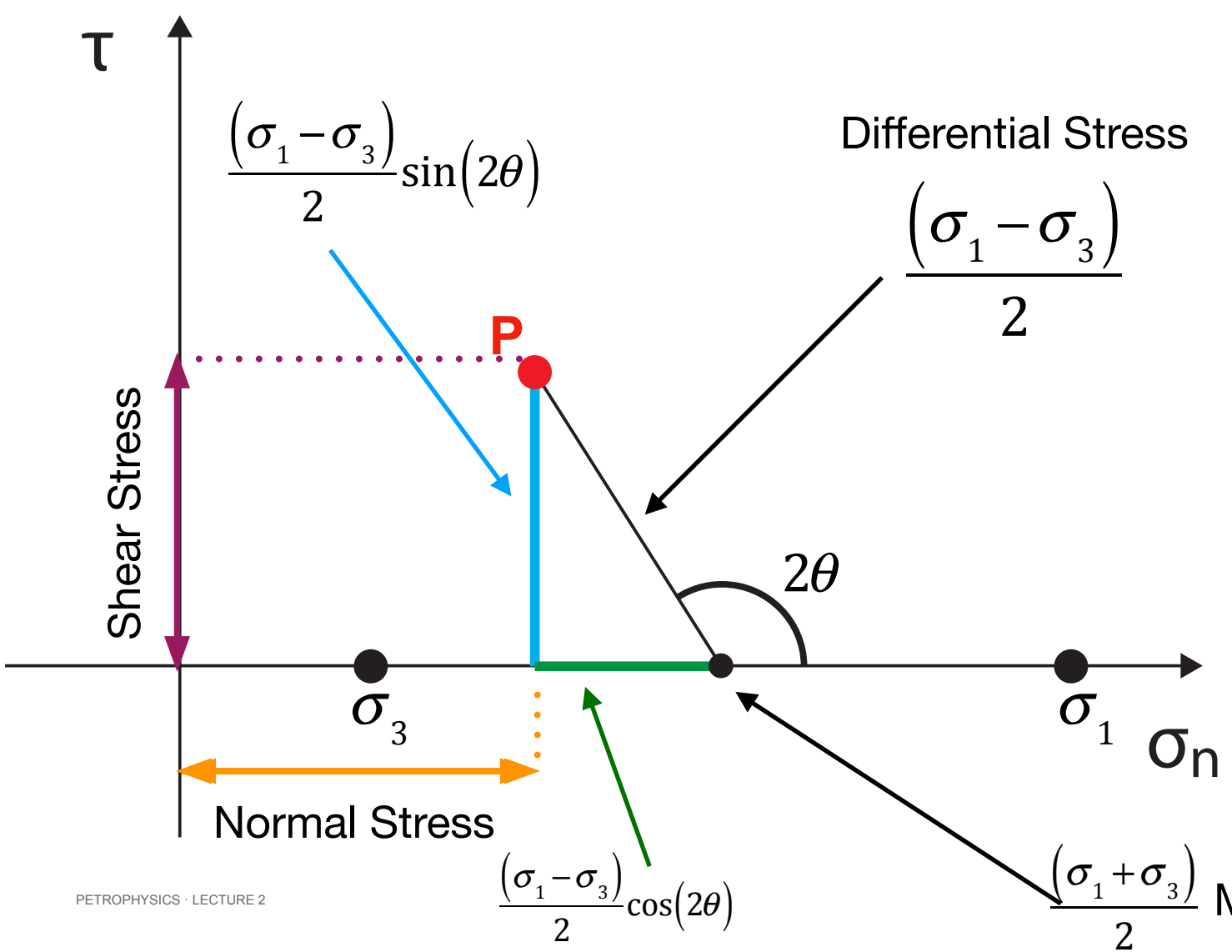
Parametric equations of the Mohr circle

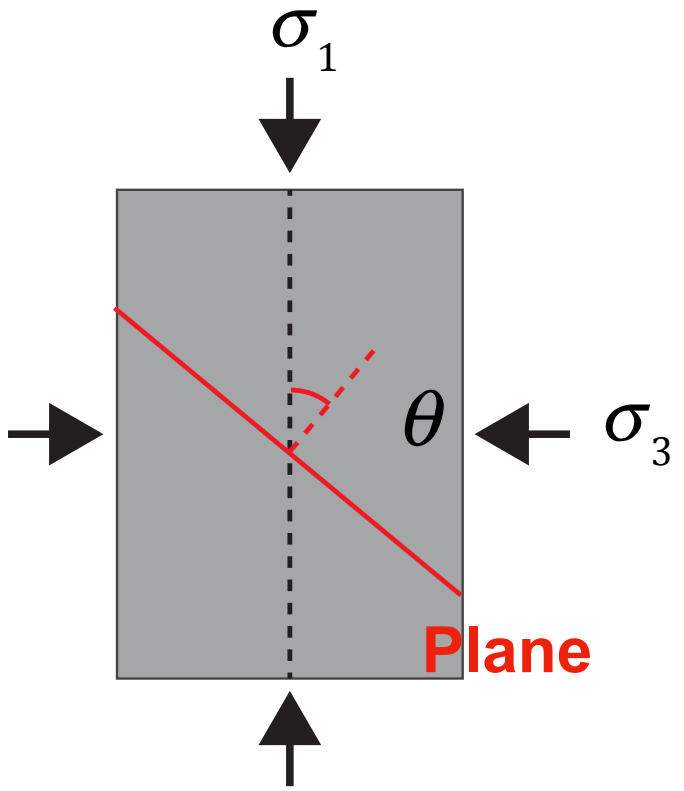
$$\sigma_n = \frac{\sigma_1 + \sigma_3}{2} - \frac{\sigma_1 - \sigma_3}{2}\cos(2\theta)$$

$$\tau = \frac{\sigma_1 - \sigma_3}{2}\sin(2\theta)$$

One plane corresponds to one point

Activity 4





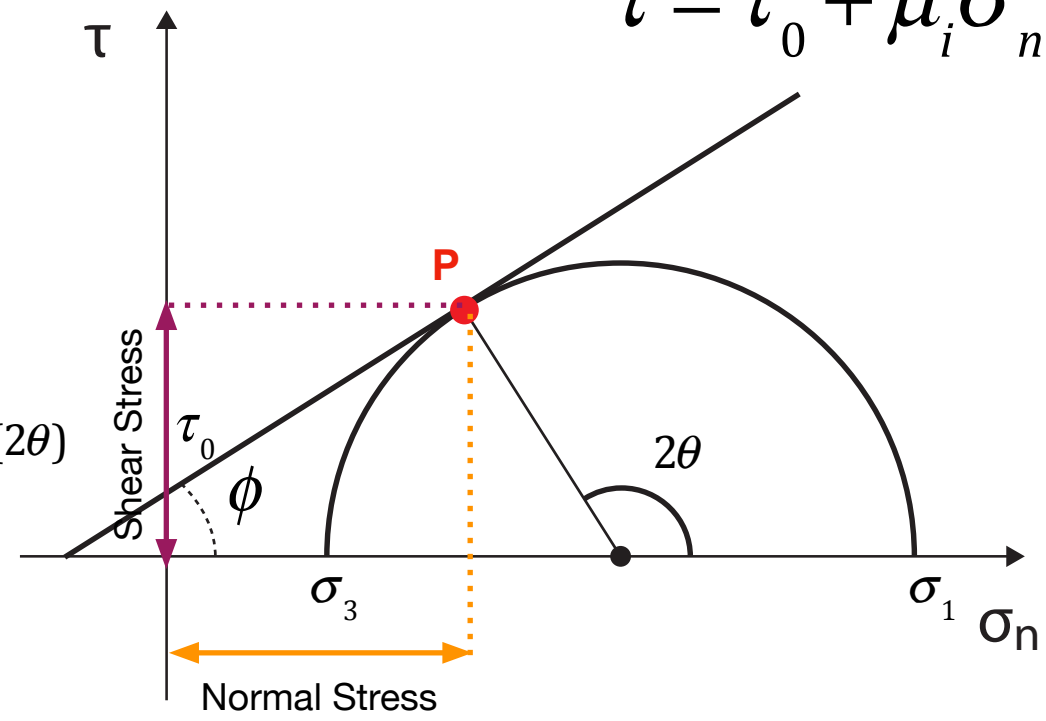
$$\sigma_n = \frac{\sigma_1+\sigma_3}{2} - \frac{\sigma_1-\sigma_3}{2}\cos(2\theta)$$

$$\tau = \frac{\sigma_1-\sigma_3}{2}\sin(2\theta)$$

Failure on a plane

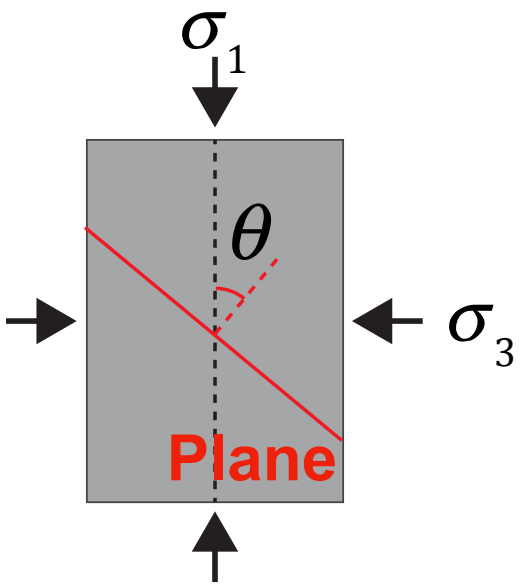
Activity 4

$$\tau = \tau_0 + \mu_i \sigma_n$$



$$\tau = \frac{\sigma_1 - \sigma_3}{2}\sin(2\theta)$$

$$\sigma_n = \frac{\sigma_1 + \sigma_3}{2} - \frac{\sigma_1 - \sigma_3}{2}\cos(2\theta)$$



$$2\theta = 90^\circ + \phi$$

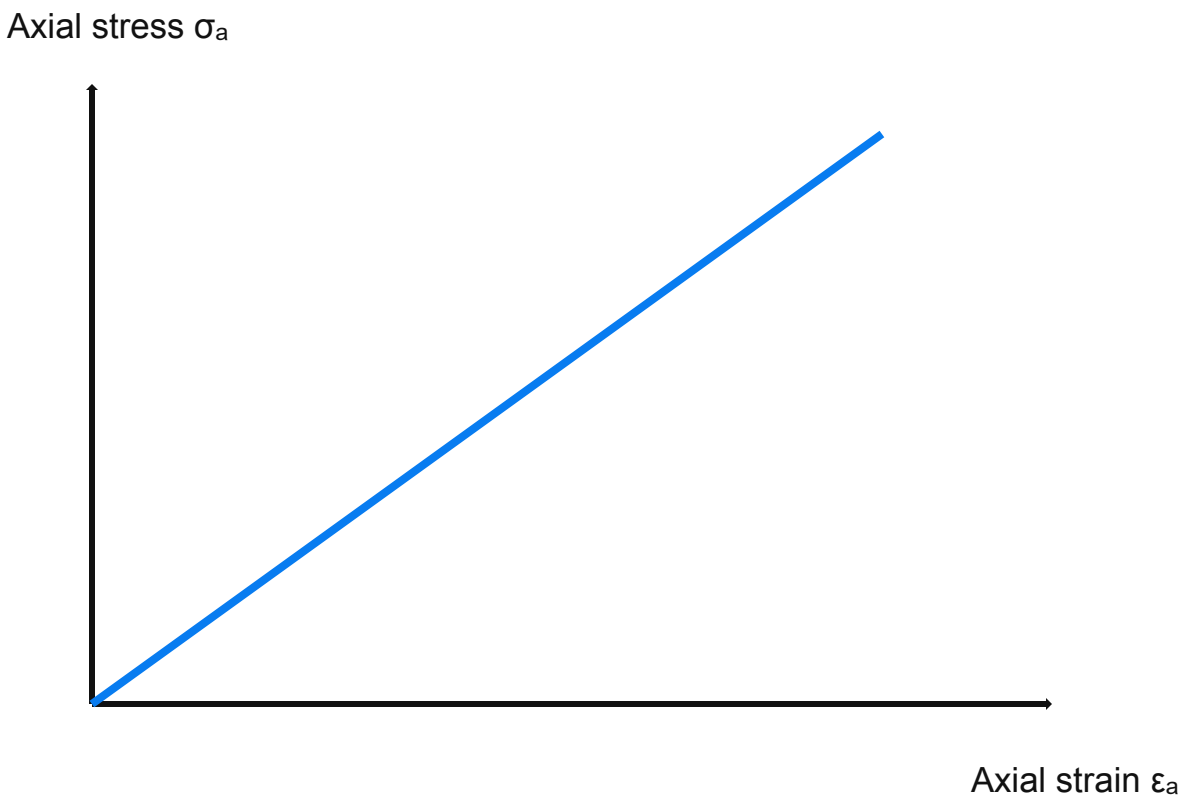
$$\theta = 45^\circ + \frac{1}{2}\phi$$

If the principal stresses are such that the circle touches the failure line, the rock will fail in shear.

The point P at which the circle is tangent to the Coulomb line represents the stress state on the plane of failure.

NEXT LECTURE

Stress and deformation become a material law



We can now describe loading and deformation.

Next

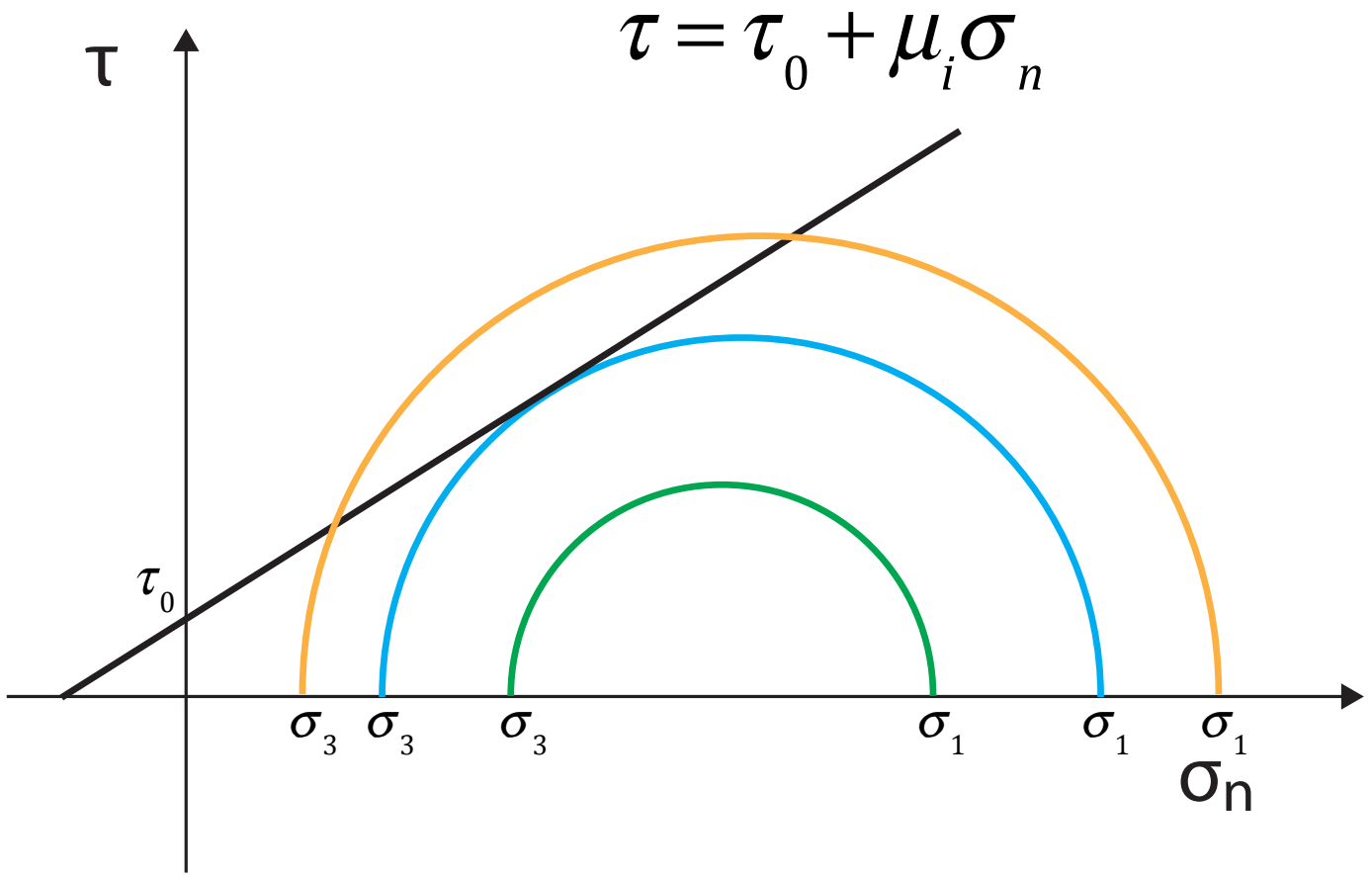
Elastic recovery
Young's modulus
Poisson's ratio

Give me feedback

Coulomb - Mohr failure analysis

Pop-Up question

Which of these stress fields will not give rise to a fault? And why?



Mohr circle construction



σ_1 = 100 MPa
σ_3 = 20 MPa

01 Mark the two intercepts

02 Find centre and radius

03 Draw the circle

04 Locate θ = 30°

What are τmax and the corresponding plane angle β?

Equal differential stress can hide different mean stress

TEST A

$\sigma_1 = 80$ MPa
$\sigma_3 = 0$ MPa

TEST B

$\sigma_1 = 100$ MPa
$\sigma_3 = 20$ MPa

Both have q = 80 MPa and R = 40 MPa

Draw both circles on the same axes. Compare their centres and the stress on a plane with θ = 60°.

Does equal differential stress imply identical loading?